JIAXI WANG

tggq61@durham.ac.uk LinkedIn
github.com/JiaxiWang05 JiaxiWang05.github.io

Research CV
Medical Robotics, Embodied AI, and Electronics Engineering
Last updated September 2026

Research Interests

Autonomous surgical robotics; vision-language-action ; embodied artificial intelligence; human–computer interaction.

Education

Electronics Engineering — **Durham University, University College, First Class Honours** — **2023–2027, UK**
Study Abroad — **University of Notre Dame, Indiana, USA** — **2025–2026 USA**

Research Experience

Student researcher under supervision of Dr Amir Atapour-Abarghouei — **ongoing**
embodied AI and vision-language-action systems — *Durham, United Kingdom*

- Cross-embodiment learning, transferable action representations, multimodal robot policies or related work on learning predictive representations for control, potentially exploration including how policies or learned representations can generalise across different robotic platforms, how VL-conditioned information can be integrated with action and state information.

Summer Undergraduate Research Intern, The Chinese University of Hong Kong — **Jun–Aug 2026**
Medical Robotics Group; Supervisor: Prof. Hongliang Ren — *Hong Kong SAR*

BiliVLA: Scene-Aware VLA for Autonomous ERCP Navigation — *Supervisor: Jinsong Lin*

- **Contributed to a system achieving 96.25% mIoU, 91.96% action precision, and 84.85% navigation success** in ERCP phantom experiments, supporting reliable target understanding and closed-loop control. Supported a two-stage pipeline combining grounding-enhanced supervised fine-tuning, GRPO reinforcement optimisation, scene-aware grounding, and safety-aware recovery; integrated hardware, multimodal trajectories, and 3-DoF actions.

Teleoperated Flexible Endoscopy Benchmark — *Supervisor: Jinsong Lin*

- **Established a reproducible multimodal benchmark pipeline** and collected **150 synchronised endoscopic video and robot-state segments**, resolving timestamp alignment across video, high-frequency robot states, and PDT hardware parameters. Built cleaning, frame-extraction, YOLO ROI-cropping, and 224 × 224 image–action trajectory workflows; deployed the open-source $\pi_{0.5}$ policy as a VLA navigation baseline.

Trileg-ESD: Vision-Guided Magnetic Soft Robot, — *Supervisor: Chi Kit Ng*

- Cooperative VLA algorithm for two soft robots. Advanced a miniature three-legged magnetic soft robot for auxiliary tissue manipulation, structuring controllable motion around squatting, leg lifting, rotation, crawling, and stance-recovery primitives. Investigated a vision-language controller mapping current and recent endoscopic frames plus motion instructions to voltage commands for three electromagnetic coils.

Google Summer of Code Mentor, International Neuroinformatics Coordinating Facility — **2026, Remote**
Open-source neuroinformatics and computational neuroscience

- Expressive Gesture to Sound Mapping Framework for GestureCap

MITACS Globalink Research Intern, McGill University, CIRMMT — **Jun–Aug 2025**
Supervisor: Prof. Suresh Krishna — *Montreal, Canada*

- **Built and evaluated a Python-based, low-latency markerless motion-to-sound workflow** using AI-assisted motion capture and Google MediaPipe for gesture-controlled music generation.
- Contributed to experimental design, coding, analysis, and reporting across engineering, neuroscience, psychology, and human–computer interaction.

HPC Summer School Student, EPCC, University of Edinburgh — **Jun–Jul 2024**
High-performance computing and scientific workflows — *Edinburgh, United Kingdom*

- Applied parallel-computing, large-scale data-processing, and scientific-computing methods to engineering problems after intensive training in HPC systems and architectures.

Publications in submission

EndoWAM: A Grounded World-Action Model for Generalisable Endoscopic Navigation, AAAI 2027 and ICLR 2027, *in submission* — **2027**

Selected Engineering Projects

Small Data, Large Models: Transfer Learning for Medical Image Classification *2026–2027, UK*

Supervised by Dr Raheleh Kafieh. Establish which pre-trained image encoder (RETFound, BioMedCLIP, ImageNet baselines) produces the most useful retinal representations under severe data limits.

Automated Material Mover *2024–2025, UK*

Supervised by Dr Aissa Ikhlef. Co-designed and prototyped a three-wheel autonomous forklift AGV with differential steering and tape-guided navigation for factory material handling; validated feasibility and ISO 3691-4 compliance through structural analysis, cost evaluation, and risk assessment; used recyclable materials and durable components for sustainability.

IMechE Repeatable Vehicle Challenge *2023–2024, UK*

Engineered, fabricated, calibrated, and repeatedly tested an autonomous vehicle designed to travel a specified distance, interact with a target, and return to a repeatable stopping position with high accuracy. The challenge focuses on designing and building a self-contained vehicle that can perform the same task repeatedly without external control.

Durham Total Artificial Heart *2024–2025, UK*

Supervised by Professor Ashraf W Khir. Developed interdisciplinary medical-engineering concepts addressing cardiovascular technology challenges and translated clinical needs into implementable engineering requirements. Defined system requirements, partitioned the wireless powering solution into subsystems, and coordinated development of high-efficiency power-transfer circuits, custom **ANSYS-designed motors**, and control electronics using **LTSpice, KiCAD and ANSYS**.

Technical Skills

Programming	Python; MATLAB; C/C++; Arduino programming; data processing and visualisation.
AI/Robotics	Computer vision; YOLO; vision-language-action models; reinforcement learning; multimodal trajectory datasets; robotic control and sensor integration.
Hardware/Tools	PCB design; analogue/digital electronics; microcontrollers; embedded systems; Multisim; LTspice; Cadence; SolidWorks; Git/GitHub; MediaPipe.
Computing	High-performance computing; parallel architectures; scientific computing; large-scale data processing and technical analysis.
Langauge	Chinese, English

Leadership and Service

Head of Design, DurHack **Sep 2024–Sep 2025**
Durham University's annual hackathon *United Kingdom*
- Led visual identity and user-facing digital communication with software and logistics teams.

Student Representative and EDI Committee Member **2023–2024**
Durham University Department of Engineering *United Kingdom*
- **Represented 150+ engineering students** and served as Engineering Society Outreach Executive.

Co-President, College Ultimate Frisbee **2024–Present**
Durham University and University of Notre Dame *UK / US*
- Managed logistics, recruitment, and multicultural team coordination; competed as a registered USA College player for Notre Dame.

President, Castle Pride **2023–2024**
University College, Durham University *United Kingdom*
- Led inclusion and community-building programmes, student discussions, and events.

Honours and Additional Information

Awards / Service	Two-time finalist, Durham University Annual Art Competition (2023, 2024); Open Day Representative, University College (2023). Pemberton award Issued by durham university, Pearson Edexcel - Certificate of Excellence, IB bilingual diploma